

Fundamentals of Cell Biology

What is a Cell?

The cell is the fundamental unit of life. All living organisms are composed of one or more cells. Cells were first discovered by Robert Hooke in 1665 when he observed thin slices of cork under a microscope. The cell theory, formulated by Schleiden and Schwann in the 1830s, states that all living things are made of cells, that the cell is the basic unit of structure and function, and that all cells arise from pre-existing cells.

Prokaryotic vs. Eukaryotic Cells

Cells are broadly classified into prokaryotic and eukaryotic types. Prokaryotic cells, found in bacteria and archaea, lack a membrane-bound nucleus; their DNA floats freely in the cytoplasm in a region called the nucleoid. Eukaryotic cells, found in plants, animals, fungi, and protists, possess a true nucleus enclosed by a double membrane, as well as various membrane-bound organelles such as mitochondria, the endoplasmic reticulum, and the Golgi apparatus.

Cell Organelles and Their Functions

The nucleus stores genetic material (DNA) and coordinates cell activities including growth, metabolism, and reproduction. Mitochondria are the powerhouses of the cell, producing ATP through cellular respiration. The endoplasmic reticulum (ER) comes in two forms: rough ER, studded with ribosomes for protein synthesis, and smooth ER, involved in lipid synthesis and detoxification. The Golgi apparatus processes, packages, and ships proteins and lipids. Lysosomes contain digestive enzymes that break down waste materials and cellular debris.

Cell Division

Cell division is essential for growth, repair, and reproduction. Mitosis is the process by which a single cell divides to produce two genetically identical daughter cells. It consists of four main phases: prophase, metaphase, anaphase, and telophase, followed by cytokinesis. Meiosis, on the other hand, is a specialized form of division that produces four genetically unique gametes (sex cells), each with half the chromosome number of the parent cell. Meiosis involves two rounds of division and introduces genetic variation through crossing over and independent assortment.

Cellular Respiration and Photosynthesis

Cellular respiration is the process by which cells convert glucose and oxygen into carbon dioxide, water, and energy (ATP). It occurs in three stages: glycolysis (in the cytoplasm), the Krebs cycle (in the mitochondrial matrix), and the electron transport chain (on the inner mitochondrial membrane). Photosynthesis, carried out by plant cells and some bacteria, captures light energy and converts carbon dioxide and water into glucose and oxygen. The light-dependent reactions occur in the thylakoid membranes, while the Calvin cycle takes place in the stroma of the chloroplast.